

MATA37H3 — Week 14

Taylor Series Applications: Binomial Series, Error Bounds & Series Gymnastics

60 minutes (Pages 1–5, self-paced) • OpenStax *Calculus Volume 2*, §6.3–§6.4 (Binomial Series; error estimation with Taylor’s Theorem; evaluating limits and nonelementary quantities with series). MAT137 Unit 14 slides (14.12–14.14: limits via series substitution; error-bounded numerical estimation) — MAT137 is the University of Toronto’s official exclusion-equivalent of MATA37H3.

KEY FACT The Binomial Series

For any real number r (not necessarily a positive integer) and $|x| < 1$,

$$(1+x)^r = \sum_{n=0}^{\infty} \binom{r}{n} x^n, \quad \binom{r}{n} = \frac{r(r-1)(r-2)\cdots(r-n+1)}{n!}, \quad \binom{r}{0} = 1.$$

When r is a nonnegative integer, $\binom{r}{n} = 0$ for $n > r$ and the series terminates — this recovers the familiar Binomial **Theorem** for finite expansions as a special case. For any other r (negative, or non-integer), the series is genuinely infinite and converges only for $|x| < 1$ (radius of convergence $R = 1$, regardless of r). The binomial series joins the six core Maclaurin series from Week 13 as a seventh standard building block.

KEY FACT Choosing the Right Error-Bound Tool

Two different tools bound the error of a truncated series, and picking the wrong one wastes effort: * **Alternating Series Remainder** (Week 10): use when the *numerical series itself* is alternating with terms decreasing to 0 — the bound is simply the first omitted term, $|R_N| \leq b_{N+1}$, no calculus needed beyond identifying the terms. * **Lagrange Remainder bound** (Theorem 6.7, Week 13): use when the series is **not** alternating, or when you need a bound on a whole *interval* of x -values at once — requires bounding $|f^{(n+1)}(t)|$ by a constant M on the interval in question.

KEY FACT Series Gymnastics for Limits and Estimation (recap)

Substituting a function’s Maclaurin series into a limit or an integral converts an indeterminate form or a nonelementary integral into ordinary term-by-term algebra. The recurring trap: truncating a series too early. Always carry the expansion **at least one order past** the terms that are explicitly being cancelled or subtracted, or the term that actually determines the answer gets thrown away by accident.

Worked Examples

EXAMPLE 1

Binomial Series, deriving $\frac{1}{\sqrt{1-x}}$

Find the first four terms of the Maclaurin series for $f(x) = \frac{1}{\sqrt{1-x}} = (1-x)^{-1/2}$.

Solution. Apply the Binomial Series with $r = -\frac{1}{2}$ and $x \rightarrow -x$: $\binom{-1/2}{0} = 1$; $\binom{-1/2}{1} = -\frac{1}{2}$; $\binom{-1/2}{2} = \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!} = \frac{3/4}{2} = \frac{3}{8}$; $\binom{-1/2}{3} = \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{3!} = \frac{-15/8}{6} = -\frac{5}{16}$. So $(1-x)^{-1/2} = 1 + (-\frac{1}{2})(-x) + \frac{3}{8}(-x)^2 + (-\frac{5}{16})(-x)^3 + \dots = 1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \dots$, valid for $|x| < 1$. Note every sign came out **positive** here because each (-1) from $\binom{r}{n}$'s alternating factors paired with each (-1) from $(-x)^n$. ■

EXAMPLE 2

limit via Maclaurin substitution, higher-order terms matter

Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$.

Solution. Substitute the full Maclaurin series: $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$. Then

$$\sin x - x + \frac{x^3}{6} = \frac{x^5}{120} - \frac{x^7}{5040} + \dots = x^5 \left(\frac{1}{120} - \frac{x^2}{5040} + \dots \right).$$

Divide by x^5 and let $x \rightarrow 0$: every remaining term has a positive power of x except the first, so the limit is $\frac{1}{120}$. *Misconception flag:* stopping the $\sin x$ expansion at x^3 (matching only the terms that are being explicitly subtracted) throws away the x^5 term that **is** the answer — always carry the series at least one order past the cancellation you expect. ■

EXAMPLE 3

comparing growth via series, a two-function contrast

Determine which is larger for small $x > 0$: $\cos(2x)$ or e^{-2x^2} , by comparing their Maclaurin series through the x^4 term.

Solution. $\cos(2x) = 1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} - \dots = 1 - 2x^2 + \frac{2}{3}x^4 - \dots$. Also $e^{-2x^2} = \sum_{n=0}^{\infty} \frac{(-2x^2)^n}{n!} =$

$1 - 2x^2 + 2x^4 - \dots$. Both start $1 - 2x^2 + \dots$, agreeing through the x^2 term, but the x^4 coefficients differ: $\frac{2}{3}$ for $\cos(2x)$ versus 2 for e^{-2x^2} . Since $\frac{2}{3} < 2$, for small $x > 0$, $\cos(2x) < e^{-2x^2}$

(concretely, $\cos(2x) - e^{-2x^2} \approx \left(\frac{2}{3} - 2\right)x^4 = -\frac{4}{3}x^4 < 0$ for small $x \neq 0$). ■

EXAMPLE 4

numerical estimation with a guaranteed error bound — Alternating Series Remainder

Estimate $B = \ln(0.9)$ to error less than 0.001.

Solution. $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ at $x = -0.1$: $\ln(0.9) = -\sum_{n=1}^{\infty} \frac{(0.1)^n}{n}$ (every term comes out negative — check directly: $n = 1$ term is -0.1 ; $n = 2$ term is -0.01 ; and so on). Treating

$b_n = \frac{(0.1)^n}{n}$ as the magnitude sequence: $b_1 = 0.1$, $b_2 = 0.005$, $b_3 = 0.000333 \dots < 0.001$. So keeping terms through $n = 2$ (partial sum $-0.1 - 0.005 = -0.105$) guarantees error $< b_3 < 0.001$. $\ln(0.9) \approx -0.105$ to the required accuracy. ■

EXAMPLE 5

a nonelementary-looking integral, estimated via series

Estimate $\int_0^1 \frac{\sin x}{x} dx$ to three decimal places (this integrand has no elementary antiderivative).

Solution. $\frac{\sin x}{x} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \dots$ (valid for all x , including

the removable point $x = 0$). Integrate term-by-term from 0 to 1: $\int_0^1 \frac{\sin x}{x} dx = 1 - \frac{1}{18} + \frac{1}{600} - \frac{1}{35280} + \dots$. This is alternating with rapidly shrinking terms; the fourth term $\frac{1}{35280} \approx 0.0000283$ is already below the needed precision, so summing the first three terms: $1 - 0.05556 + 0.001667 \approx 0.94611$. $\int_0^1 \frac{\sin x}{x} dx \approx 0.946$. ■

EXAMPLE 6

error bound with the Lagrange Remainder — non-alternating case

Estimate $e^{0.2}$ using $p_3(x)$ for e^x at $a = 0$, and give a guaranteed bound on the error using the Lagrange Remainder (Theorem 6.7).

Solution. $p_3(0.2) = 1 + 0.2 + \frac{0.2^2}{2} + \frac{0.2^3}{6} = 1 + 0.2 + 0.02 + 0.001\bar{3} = 1.221\bar{3}$. Since e^x 's series has **all positive terms** at $x = 0.2 > 0$, the Alternating Series Remainder does not apply — use the Lagrange bound instead: $f^{(4)}(t) = e^t$, and on $[0, 0.2]$, $e^t \leq e^{0.2} < e^{0.3} \approx 1.35$ (a safe, easy-to-justify overestimate for M without needing the answer itself). So

$$|R_3(0.2)| \leq \frac{M}{4!}(0.2)^4 \leq \frac{1.35}{24}(0.0016) \approx 0.00009.$$

So $e^{0.2} \approx 1.2213$ with guaranteed error under 0.0001 (matches the true value $e^{0.2} \approx 1.22140$). ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

Binomial Series

1. Find the first four terms of the Maclaurin series for $f(x) = (1+x)^{-2}$ using the Binomial Series with $r = -2$, and check your answer against Week 11's result $\frac{1}{(1-x)^2} =$

$\sum (n+1)x^n$ (substitute $x \rightarrow -x$ to compare).

HINT Use the general binomial coefficient formula $\binom{r}{n} = \frac{r(r-1)(r-2)\cdots(r-n+1)}{n!}$ with $r = -2$, and simplify the growing product of negative factors before multiplying by x^n .

CAUTION — before you solve Cross-check your first few terms against Week 11 Problem 6's series for $\frac{1}{(1-x)^2} = \sum (n+1)x^n$ by substituting $x \rightarrow -x$ — the two derivations should agree term by term.

Your working:

2. Find the first three nonzero terms of the Maclaurin series for $f(x) = \sqrt{1+x} = (1+x)^{1/2}$.

HINT Apply the Binomial Series with $r = \frac{1}{2}$, computing $\binom{1/2}{0}, \binom{1/2}{1}, \binom{1/2}{2}, \binom{1/2}{3}$ in turn.

CAUTION — before you solve $\binom{1/2}{n}$ does not simplify the same way $\binom{-1/2}{n}$ does — recompute each coefficient from the formula rather than reusing the signs from the $(1-x)^{-1/2}$ example.

Your working:

Limits and Growth Comparisons

3. Evaluate $\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4}$ using Maclaurin series.

HINT Substitute the Maclaurin series for $\cos x$, subtract $1 - x^2/2$ term by term, then divide the remaining series by x^4 .

CAUTION — before you solve The lowest surviving power after subtracting $1 - x^2/2$ is x^4 , not x^6 — double-check which term of the $\cos x$ series you're dividing by x^4 .

Your working:

Numerical Estimation

4. Estimate $\sin(1)$ using the first three nonzero terms of its Maclaurin series, and use the Alternating Series Remainder to state a guaranteed error bound on your estimate.

HINT Write $\sin(1)$ as the alternating series $\sum \frac{(-1)^n}{(2n+1)!}$ evaluated at $x = 1$, then find the smallest n making the next term's magnitude fall under the required tolerance.

CAUTION — before you solve The Alternating Series Remainder bound only applies once the terms are genuinely decreasing in magnitude and alternating in sign — confirm both before trusting the bound.

Your working:

5. Estimate $\int_0^1 e^{-x^2} dx$ to two decimal places using the Maclaurin series for e^{-x^2} , integrated term-by-term (state how many terms you needed and why you can stop there).

HINT Substitute the Maclaurin series for e^{-x^2} (substitute $-x^2$ into the e^x series) and integrate term by term over $[0, 1]$.

CAUTION — before you solve This series has no alternating-sign guarantee issue here since the signs do alternate — but make sure you integrate x^{2n} to $\frac{x^{2n+1}}{2n+1}$ correctly before evaluating at the bounds.

Your working:

6. Estimate $\cos(0.3)$ using $p_2(x)$ for $\cos x$ at $a = 0$, and use the Lagrange Remainder to give a guaranteed error bound (use $M = 1$, since every derivative of $\cos x$ is bounded by 1).

HINT Bound $|f^{(4)}(t)| = |\cos t| \leq 1$ (or use whichever derivative order matches your chosen degree) on the interval between 0 and 0.3, then plug into the Lagrange Remainder formula.

CAUTION — before you solve Since $|\cos t| \leq 1$ everywhere, you can safely use $M = 1$ regardless of the interval — don't overcomplicate the bound by trying to find a tighter M .

Your working:

Reasoning

7. A student claims: “Since the Taylor series of f at a converges for all $x \in \mathbb{R}$, we know f equals its Taylor series everywhere.” Explain precisely why this claim is **false** in general, citing the distinction between “the series converges” and “the series converges to f ” (Theorem 6.8, Week 13), and what extra ingredient (the Lagrange remainder bound) is needed to upgrade convergence to equality.

HINT Recall Theorem 6.8’s exact requirement: a Taylor series converges to f only where the Lagrange Remainder $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ — convergence of the series alone doesn’t guarantee this.

CAUTION — before you solve The classic counterexample here is a function whose Taylor series converges everywhere but only equals the function at the center — don’t assume ‘the series converges’ and ‘the series equals f ’ are the same claim.

Your working:
